

Sales Prediction Using Machine Learning

Name: Nuvvula Anuradha

Internship Domain: Data Science

Organization: CodeAlpha

Project Title: Sales Prediction Using Machine Learning

Abstract

The Sales Prediction project uses advertising expenditure data to predict future sales. A Linear Regression model is trained using TV, Radio, and Newspaper advertising budgets to estimate sales performance.

Introduction

Sales forecasting is an important business activity that helps organizations plan marketing strategies and allocate resources effectively. Machine learning techniques can improve forecasting accuracy by analyzing historical advertising and sales data.

Objective

The objective of this project is to build a predictive model that estimates sales based on advertising investments across multiple media channels.

Dataset Description

The dataset contains the following attributes:

- TV Advertising Budget
- Radio Advertising Budget
- Newspaper Advertising Budget
- Sales Revenue

The dataset consists of historical advertising expenditures and corresponding sales figures.

Tools and Technologies Used

- Python
- Pandas
- NumPy

- Scikit-Learn
- Visual Studio Code

Methodology

1. Load the advertising dataset.
2. Explore and analyze the data.
3. Select advertising budgets as input features.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Evaluate model performance using the model score.

Results

The model achieved a score of approximately 0.90, indicating strong predictive performance. The results show a significant relationship between advertising expenditure and sales outcomes.

Conclusion

This project demonstrates how machine learning can be used for sales forecasting and business decision-making. Advertising investments play an important role in influencing sales performance.

Future Scope

- Include additional marketing channels.
- Use advanced forecasting algorithms.
- Build an interactive sales prediction dashboard.

